

Project 1
CSE 464
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Project Description

This project implements a graph parser and manager that processes graphs defined in DOT format. The application is designed to:

- Parse a DOT file to create a directed graph object.
- Add nodes and edges to the graph.
- Output the graph details as text.
- Export the graph in DOT format and generate a graphical output (e.g., PNG) using Graphviz.

Prerequisites

- Java JDK 11 or above
- Maven 3.6+
- Graphviz installed and available in your system PATH
- Git

Instructions to run:

Clone this repo <https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10.git>

Navigate to this project directory : CSE-464-2025-gmehta10

Run this command to compile and test : mvn clean package

Then go to the main file and run it using the run button in IntelliJ app

Feature 1:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/commit/9a5d2110522e6dce1ba2bcb4742daff8035e36aa>

```
/Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -javaagent:/Applications/IntelliJ IDEA CE.app/Contents/lib/idea_rt.jar=59532
Number of nodes: 8
Nodes: [a, b, c, d, e, f, g, h]
Number of edges: 9
Edges:
a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h

Process finished with exit code 0
```

Feature 2:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/commit/ee7516e5e23fdb0f85a11935c6ed1c27bd1b5d89>

Adding nodes...

Node 'X' added.

Updated Graph:

Number of nodes: 11

Nodes: [a, b, c, d, e, f, g, h, X, Y, Z]

Number of edges: 9

Edges:

a -> b

b -> c

c -> d

d -> a

a -> e

e -> f

e -> g

f -> h

g -> h

Feature 3:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/commit/68e4419b1e9a8cb522f9defa1f0103f110be29eb>

Adding edges...

Edge 'X -> Y' added.

Edge 'B -> C' added.

Edge 'X -> Y' already exists.

Updated Graph:

Number of nodes: 13

Nodes: [a, b, B, c, C, d, e, f, g, h, X, Y, Z]

Number of edges: 11

Edges:

a -> b

b -> c

c -> d

d -> a

a -> e

e -> f

e -> g

f -> h

g -> h

X -> Y

B -> C

Feature 4:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/commit/6a31b7d49596655bd2617ddab13ad3ca9459d44b>

```
gautammehta@Gautams-MacBook-Air-2 PJ1 % mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:GraphParserProject >-----
[INFO] Building GraphParserProject 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ GraphParserProject ---
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-11/zstd-jni-1.5.5-11.jar (6.8 MB at 21 MB/s)
[INFO] Building jar: /Users/gautammehta/Desktop/CSE-464-2025-gmehta10/PJ1/target/GraphParserProject-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.604 s
[INFO] Finished at: 2025-03-04T17:21:07-07:00
[INFO] -----
```

